

Explicit Reconstructed 2-RDM Elements and Full Derivation of the Relativistic PNOF5, PNOF7, PNOF7s, and GNOF Energy Functionals

Abstract

Starting from the general no-pair relativistic 2-RDM reconstruction of Appendix D of SciPost Chem. **1**, 004 (2022) [Eqs. (97a)–(98f)], we write down, fully explicitly and separately for the relativistic PNOF5, PNOF7, PNOF7s, and GNOF functionals, every 2-RDM element ${}^2D_{ij}^{kl}$ entering the natural-orbital energy expression Eq. (64) of that paper. We then substitute these explicit elements into Eq. (64), evaluate every one of its six blocks in full detail, and show that the result is identical, term by term, to the published energy expressions Eqs. (72)–(76) of Sec. 3.5.3. The no-pair Dirac–Hartree–Fock limit is recovered as a corollary.

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1 Scope, conventions, and notation

We work throughout within the no-pair approximation and impose Kramers’ pairing symmetry, restricting attention to ‘spin-compensated’ systems, exactly as in Sec. 3.5 of the source paper. Natural spinors are grouped into Kramers pairs $(i, \bar{i})$, with i running over ‘half of the positive-energy spinors’ and $\bar{i}$ its time-reversed partner. Spin compensation means

$$n_{\bar{i}} = n_i \quad \text{for all } i. \quad (1)$$

The spinor space is partitioned into mutually disjoint subspaces (geminals),

$$\Omega = \bigcup_p \Omega_p, \quad \Omega_p \cap \Omega_q = \emptyset \ (p \neq q), \quad \sum_p n_p = \frac{1}{2} N_e, \quad (2)$$

with $n_p \equiv n_i$ for the reference (‘principal’) orbital of Ω_p and $h_p = 1 - n_p$.

We use the same integral symbols as the source paper,

$$h_{ii} = \int d\mathbf{r} d\mathbf{r}' \tilde{\chi}_i^\dagger(\mathbf{r}') [\delta(\mathbf{r} - \mathbf{r}') \hat{T}_D(\mathbf{r}) + v_{\text{ext}}^{\text{nl}}(\mathbf{r}, \mathbf{r}')] \tilde{\chi}_i(\mathbf{r}), \quad (3)$$

$$J_{ij} = \langle ij | ij \rangle, \quad J_{ij}^G = \langle ij | \boldsymbol{\alpha} \cdot \boldsymbol{\alpha} | ij \rangle, \quad (4)$$

$$K_{ij} = \langle ij | j\bar{i} \rangle, \quad K_{ij}^G = \langle ij | \boldsymbol{\alpha} \cdot \boldsymbol{\alpha} | j\bar{i} \rangle, \quad (5)$$

$$L_{ij} = \langle \bar{i}j | j\bar{i} \rangle, \quad L_{ij}^G = \langle \bar{i}j | \boldsymbol{\alpha} \cdot \boldsymbol{\alpha} | j\bar{i} \rangle, \quad (6)$$

with the paper’s stated property $L_{ii} = 0$ (while $L_{ii}^G \neq 0$ in general).

The cumulant reconstruction of Appendix D uses two auxiliary real, symmetric matrices, Δ_{ij} and $\Pi_{ij} \equiv \Pi_{i\bar{i},j\bar{j}}$ (the four possible bar-orderings of the latter are all numerically equal, as shown explicitly below Eq. (97e) of the source paper). Following the prescription used to build Eqs. (72)–(76), we adopt

$$\Delta_{ij} = \Delta_{i\bar{j}} = \Delta_{\bar{i}j} = \Delta_{\bar{i}\bar{j}}, \quad \Delta_{ii} = n_i^2, \quad \Pi_{ii} = n_i, \quad (7)$$

and, for the off-diagonal part of Δ , the subspace ansatz

$$\Delta_{ij} = \begin{cases} n_i n_j, & i, j \in \Omega_p \ (i \neq j), \\ 0, & i \in \Omega_q, \ j \in \Omega_p, \ p \neq q. \end{cases} \quad (8)$$

This ansatz is the *same for all four functionals*; only Π^{inter} differs between PNOF5, PNOF7, PNOF7s, and GNOF. For compactness define

$$A_{ij} \equiv \frac{n_i n_j - \Delta_{ij}}{2} \implies A_{ij} = \begin{cases} 0, & i, j \in \Omega_p, \\ \frac{n_i n_j}{2}, & i \in \Omega_q, \ j \in \Omega_p \ (p \neq q). \end{cases} \quad (9)$$

The intra-subspace part of Π is likewise common to all four functionals [Eq. (74) of the source paper]:

$$\Pi_{ij}^{\text{intra}} = \begin{cases} -\sqrt{n_i n_j}, & i = p \text{ or } j = p \text{ (principal orbital of } \Omega_p), \\ +\sqrt{n_i n_j}, & \text{otherwise,} \end{cases} \quad i, j \in \Omega_p, \ i \neq j. \quad (10)$$

The inter-subspace part, $\Pi_{ij;pq}^{\text{inter}}$ for $i \in \Omega_q, j \in \Omega_p, p \neq q$, is what distinguishes the four functionals (Table 3 of the source paper) and is given explicitly in Sec. 3 below.

2 The general reconstructed 2-RDM [Eqs. (98a)–(98f)]

For reference, the general (functional-independent) reconstruction of Appendix D reads

$${}^2D_{ij}^{kl} = \frac{n_i n_j - \Delta_{ij}}{2} \delta_{ik} \delta_{jl} - \frac{n_i n_j - \Delta_{ij}}{2} \delta_{il} \delta_{jk}, \quad (11)$$

$${}^2D_{ij}^{\bar{k}\bar{l}} = \frac{n_{\bar{i}} n_{\bar{j}} - \Delta_{\bar{i}\bar{j}}}{2} \delta_{ik} \delta_{jl} - \frac{n_{\bar{i}} n_{\bar{j}} - \Delta_{\bar{i}\bar{j}}}{2} \delta_{il} \delta_{jk}, \quad (12)$$

$${}^2D_{ij}^{k\bar{l}} = \frac{n_i n_{\bar{j}} - \Delta_{i\bar{j}}}{2} \delta_{ik} \delta_{jl} + \frac{\Pi_{i\bar{i}, k\bar{k}}}{2} \delta_{ij} \delta_{kl}, \quad (13)$$

$${}^2D_{ji}^{\bar{l}k} = \frac{n_i n_{\bar{j}} - \Delta_{\bar{j}i}}{2} \delta_{ik} \delta_{jl} + \frac{\Pi_{\bar{i}i, \bar{k}k}}{2} \delta_{ij} \delta_{kl}, \quad (14)$$

$${}^2D_{ji}^{k\bar{l}} = -\frac{n_i n_{\bar{j}} - \Delta_{\bar{j}i}}{2} \delta_{ik} \delta_{jl} - \frac{\Pi_{i\bar{i}, k\bar{k}}}{2} \delta_{ij} \delta_{kl}, \quad (15)$$

$${}^2D_{ij}^{\bar{l}k} = -\frac{n_i n_{\bar{j}} - \Delta_{i\bar{j}}}{2} \delta_{ik} \delta_{jl} - \frac{\Pi_{i\bar{i}, \bar{k}k}}{2} \delta_{ij} \delta_{kl}. \quad (16)$$

Using spin compensation (1), the Δ symmetry (7), and the equality of all bar-orderings of Π , every element that enters Eq. (64) reduces to one of only two closed forms, valid for *any* i, k (equal or not):

$${}^2D_{ii}^{k\bar{k}} = \frac{n_i^2 - \Delta_{ii}}{2} \delta_{ik} + \frac{\Pi_{ik}}{2} = \frac{\Pi_{ik}}{2}, \quad (17)$$

$${}^2D_{kk}^{\bar{i}\bar{i}} = -\frac{n_k^2 - \Delta_{kk}}{2} \delta_{ik} - \frac{\Pi_{ik}}{2} = -\frac{\Pi_{ik}}{2}, \quad (18)$$

where the diagonal- Δ convention $\Delta_{ii} = n_i^2$ kills the first term identically (this is what makes the diagonal case, $i = k$, consistent with the off-diagonal one via $\Pi_{ii} = n_i$). Equations (17)–(18) are the compact ‘Eq. 98’ library used repeatedly below.

The remaining ‘Coulomb/exchange-type’ elements needed are, for $i \neq j$,

$${}^2D_{ij}^{ij} = {}^2D_{ij}^{\bar{i}\bar{j}} = {}^2D_{ij}^{i\bar{j}} = {}^2D_{ij}^{\bar{i}j} = A_{ij}, \quad {}^2D_{ij}^{ji} = {}^2D_{ji}^{\bar{j}\bar{i}} = -A_{ij}, \quad (19)$$

and the four ‘spin-flip’ (L -type) elements, for $i \neq j$,

$${}^2D_{ij}^{\bar{j}\bar{i}} = {}^2D_{ji}^{\bar{i}j} = {}^2D_{ji}^{\bar{j}i} = {}^2D_{ij}^{\bar{i}j} = -A_{ij}. \quad (20)$$

For $i = j$ the pure-unbarred/pure-barred elements ${}^2D_{ii}^{ii} = {}^2D_{ii}^{\bar{i}\bar{i}} = 0$ vanish identically by antisymmetry, while ${}^2D_{ii}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{i}i} = n_i/2$ follow from (17) at $i = k$, and ${}^2D_{ii}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{i}i} = -n_i/2$ follow from (18) at $i = k$.

3 Explicit 2-RDM elements for each functional

Because (9), (19), and (20) depend on Δ only, they are *identical for all four functionals* once the intra/inter split (8) is applied. The functional identity enters only through Π_{ik} in (17)–(18), via Π^{intra} [Eq. (10), common to all] and Π^{inter} [functional-dependent]. We now list, functional by functional, the complete set of explicit 2-RDM elements.

Common intra-subspace elements ($i, j \in \Omega_p$, $i \neq j$)

$${}^2D_{ij}^{ij} = {}^2D_{ij}^{\bar{i}\bar{j}} = {}^2D_{ij}^{\bar{i}j} = {}^2D_{ij}^{i\bar{j}} = 0, \quad {}^2D_{ij}^{ji} = {}^2D_{ji}^{\bar{i}\bar{j}} = 0, \quad (21)$$

$${}^2D_{ij}^{j\bar{i}} = {}^2D_{ji}^{\bar{j}i} = {}^2D_{ji}^{\bar{j}\bar{i}} = {}^2D_{ji}^{j\bar{i}} = 0, \quad (22)$$

$${}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{i}} = \frac{\Pi_{ij}^{\text{intra}}}{2} = \mp \frac{\sqrt{n_i n_j}}{2}, \quad (23)$$

$${}^2D_{jj}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{j}\bar{j}} = {}^2D_{jj}^{\bar{i}\bar{j}} = {}^2D_{ii}^{\bar{j}j} = -\frac{\Pi_{ij}^{\text{intra}}}{2} = \pm \frac{\sqrt{n_i n_j}}{2}, \quad (24)$$

(upper sign if i or j is the principal orbital of Ω_p , lower sign otherwise), all vanishing whenever $\Delta_{ij} = n_i n_j$ makes $A_{ij} = 0$, since intra-subspace pairs never contribute a Coulomb/exchange-type or L -flip-type 2-RDM element – only the Π^{intra} pairing term survives.

Common diagonal elements ($i = j$, any functional)

$${}^2D_{ii}^{ii} = {}^2D_{ii}^{\bar{i}\bar{i}} = 0, \quad {}^2D_{ii}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{i}i} = \frac{n_i}{2}, \quad {}^2D_{ii}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{i}i} = -\frac{n_i}{2}. \quad (25)$$

Inter-subspace elements ($i \in \Omega_q$, $j \in \Omega_p$, $q \neq p$)

Here $A_{ij} = n_i n_j / 2$, so

$${}^2D_{ij}^{ij} = {}^2D_{ij}^{\bar{i}\bar{j}} = {}^2D_{ij}^{\bar{i}j} = {}^2D_{ij}^{i\bar{j}} = \frac{n_i n_j}{2}, \quad {}^2D_{ij}^{ji} = {}^2D_{ji}^{\bar{i}\bar{j}} = -\frac{n_i n_j}{2}, \quad (26)$$

$${}^2D_{ij}^{j\bar{i}} = {}^2D_{ji}^{\bar{j}i} = {}^2D_{ji}^{\bar{j}\bar{i}} = {}^2D_{ji}^{j\bar{i}} = -\frac{n_i n_j}{2}, \quad (27)$$

$${}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{i}} = \frac{\Pi_{ij;pq}^{\text{inter}}}{2}, \quad (28)$$

$${}^2D_{jj}^{\bar{i}\bar{i}} = {}^2D_{ii}^{jj} = {}^2D_{ii}^{\bar{j}\bar{j}} = {}^2D_{jj}^{\bar{j}j} = -\frac{\Pi_{ij;pq}^{\text{inter}}}{2}. \quad (29)$$

These are common in *form* to all four functionals; only the value of $\Pi_{ij;pq}^{\text{inter}}$ in (28)–(29) changes. We now give this value, and hence the fully explicit inter-subspace 2-RDM elements, functional by functional.

3.0.1 PNOF5

$$\Pi_{ij;pq}^{\text{inter, PNOF5}} = 0 \implies {}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{i}} = {}^2D_{jj}^{\bar{i}\bar{j}} = {}^2D_{ii}^{\bar{j}\bar{j}} = {}^2D_{jj}^{\bar{i}i} = {}^2D_{ii}^{\bar{j}j} = 0. \quad (30)$$

For PNOF5 the inter-subspace 2-RDM therefore contains *only* the ordinary Coulomb/exchange/ L -flip elements (26)–(27); there is no inter-subspace pairing contribution.

3.0.2 PNOF7

$$\Pi_{ij;pq}^{\text{inter, PNOF7}} = -\sqrt{n_i n_j h_i h_j} \implies \begin{aligned} {}^2D_{ii}^{jj} &= {}^2D_{jj}^{\bar{i}\bar{i}} = -\frac{1}{2}\sqrt{n_i n_j h_i h_j}, \\ {}^2D_{jj}^{\bar{i}\bar{i}} &= {}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{j}} = {}^2D_{ii}^{\bar{j}\bar{j}} = +\frac{1}{2}\sqrt{n_i n_j h_i h_j}. \end{aligned} \quad (31)$$

3.0.3 PNOF7s

$$\Pi_{ij;pq}^{\text{inter, PNOF7s}} = -4n_i n_j h_i h_j \implies \begin{aligned} {}^2D_{ii}^{jj} &= {}^2D_{jj}^{\bar{i}\bar{i}} = -2n_i n_j h_i h_j, \\ {}^2D_{jj}^{\bar{i}\bar{i}} &= {}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{j}} = {}^2D_{ii}^{\bar{j}\bar{j}} = +2n_i n_j h_i h_j. \end{aligned} \quad (32)$$

3.0.4 GNOF

Define, with $h_p = 1 - n_p$ (the hole of the principal orbital of Ω_p),

$$n_i^d = n_i \frac{h_p^d}{h_p}, \quad h_p^d = h_p \exp \left[- \left(\frac{h_p}{0.02\sqrt{2}} \right)^2 \right]. \quad (33)$$

Then, for $i \in \Omega_q$, $j \in \Omega_p$, $q \neq p$,

$$\Pi_{ij;pq}^{\text{inter, GNOF}} = \begin{cases} n_i^d n_j^d - \sqrt{n_i n_j h_i h_j} - \sqrt{n_i^d n_j^d}, & i > q, j = p \text{ or } i = q, j > p, \\ n_i^d n_j^d - \sqrt{n_i n_j h_i h_j} + \sqrt{n_i^d n_j^d}, & i > q, j > p, \\ 0, & i = q, j = p, \end{cases} \quad (34)$$

so that the fully explicit GNOF inter-subspace elements are

$${}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{i}} = \frac{\Pi_{ij;pq}^{\text{inter, GNOF}}}{2}, \quad {}^2D_{jj}^{\bar{i}\bar{i}} = {}^2D_{ii}^{jj} = {}^2D_{jj}^{\bar{i}\bar{i}} = {}^2D_{ii}^{\bar{j}\bar{j}} = -\frac{\Pi_{ij;pq}^{\text{inter, GNOF}}}{2}, \quad (35)$$

with $\Pi_{ij;pq}^{\text{inter, GNOF}}$ given piecewise by (34).

Equations (30), (31), (32), and (35) – together with the common elements (21)–(25) and (26)–(27) – constitute the complete, explicit reconstructed 2-RDM (‘Eq. 98’) for each of the four functionals. We now show, step by step, that inserting these elements into Eq. (64) reproduces Eqs. (72)–(76).

4 The energy expression, Eq. (64)

The np-ReNOFT energy under Kramers symmetry (Eq. (64) of the source paper) is

$$\begin{aligned} E = & \sum_i h_{ii}(n_i + n_{\bar{i}}) \\ & + \sum_{i,j} \left({}^2D_{ij}^{ij} + {}^2D_{ij}^{\bar{i}\bar{j}} + {}^2D_{ij}^{\bar{j}\bar{i}} + {}^2D_{ij}^{\bar{i}\bar{j}} \right) J_{ij} \\ & - \sum_{i,j} \left({}^2D_{ij}^{ij} - {}^2D_{ij}^{\bar{i}\bar{j}} - {}^2D_{ij}^{\bar{j}\bar{i}} + {}^2D_{ij}^{\bar{i}\bar{j}} \right) J_{ij}^G \\ & + \sum_{i,j} \left[{}^2D_{ij}^{ji} + {}^2D_{ji}^{\bar{i}\bar{j}} \right] (K_{ij} - K_{ij}^G) \\ & + \frac{1}{2} \sum_{i,j} \left[{}^2D_{ij}^{\bar{j}\bar{i}} + {}^2D_{ji}^{\bar{i}\bar{j}} + {}^2D_{ji}^{\bar{j}\bar{i}} + {}^2D_{ij}^{\bar{j}\bar{i}} \right] (L_{ij} - L_{ij}^G) \\ & + \sum_{i \neq j} \left[{}^2D_{ii}^{jj} + {}^2D_{jj}^{\bar{i}\bar{i}} \right] (K_{ij} + K_{ij}^G) \\ & - \frac{1}{2} \sum_{i \neq j} \left[{}^2D_{jj}^{\bar{i}\bar{i}} + {}^2D_{ii}^{jj} + {}^2D_{jj}^{\bar{i}\bar{i}} + {}^2D_{ii}^{jj} \right] (L_{ij} + L_{ij}^G). \end{aligned} \quad (36)$$

The six sums (call them, in order, the h -, J -, J^G -, K -, L -, ΠK -, and ΠL -blocks) are evaluated one at a time below, generically in A_{ij} and Π_{ij} ; the functional identity is inserted only at the very end.

4.1 The h -block

By spin compensation (1),

$$\sum_i h_{ii}(n_i + n_{\bar{i}}) = 2 \sum_i n_i h_{ii}. \quad (37)$$

4.2 The Coulomb (J) block

Define $C_{ij}^J = {}^2D_{ij}^{ij} + {}^2D_{ij}^{\bar{i}\bar{j}} + {}^2D_{ij}^{i\bar{j}} + {}^2D_{ij}^{\bar{i}j}$. For $i \neq j$, using (19),

$$C_{ij}^J = 4A_{ij} = 2(n_i n_j - \Delta_{ij}). \quad (38)$$

For $i = j$, using (25),

$$C_{ii}^J = 0 + 0 + \frac{n_i}{2} + \frac{n_i}{2} = n_i. \quad (39)$$

Hence

$$E_J = \sum_i n_i J_{ii} + \sum_{i \neq j} 2(n_i n_j - \Delta_{ij}) J_{ij}. \quad (40)$$

4.3 The Gaunt (J^G) block

Define $C_{ij}^{J^G} = {}^2D_{ij}^{ij} - {}^2D_{ij}^{\bar{i}\bar{j}} - {}^2D_{ij}^{i\bar{j}} + {}^2D_{ij}^{\bar{i}j}$. For $i \neq j$, all four elements equal A_{ij} , so

$$C_{ij}^{J^G} = A_{ij} - A_{ij} - A_{ij} + A_{ij} = 0. \quad (41)$$

For $i = j$,

$$C_{ii}^{J^G} = 0 - \frac{n_i}{2} - \frac{n_i}{2} + 0 = -n_i. \quad (42)$$

Since Eq. (36) contains $-C_{ij}^{J^G} J_{ij}^G$,

$$E_{J^G} = \sum_i n_i J_{ii}^G. \quad (43)$$

No inter- or intra-subspace ($i \neq j$) Gaunt-Coulomb term survives: this is why Eqs. (73) and (75) contain J_{ii}^G (diagonal only) and never J_{ij}^G ($i \neq j$).

4.4 The exchange (K) block

Define $C_{ij}^K = {}^2D_{ij}^{ji} + {}^2D_{ji}^{\bar{i}\bar{j}}$. For $i \neq j$, using (19),

$$C_{ij}^K = -2A_{ij} = -(n_i n_j - \Delta_{ij}). \quad (44)$$

For $i = j$ both terms vanish by antisymmetry. Hence

$$E_K = - \sum_{i \neq j} (n_i n_j - \Delta_{ij}) (K_{ij} - K_{ij}^G). \quad (45)$$

4.5 The spin-flip (L) block

Define $C_{ij}^L = {}^2D_{ij}^{j\bar{i}} + {}^2D_{ji}^{\bar{i}\bar{j}} + {}^2D_{ji}^{\bar{i}j} + {}^2D_{ij}^{\bar{j}i}$ (note that in Eq. (36) this sum runs over *all* i, j , including $i = j$, unlike the K - and Π -blocks).

For $i \neq j$, using (20),

$$C_{ij}^L = -4A_{ij} = -2(n_i n_j - \Delta_{ij}). \quad (46)$$

With the $\frac{1}{2}$ prefactor of Eq. (36),

$$E_L^{i \neq j} = - \sum_{i \neq j} (n_i n_j - \Delta_{ij}) (L_{ij} - L_{ij}^G). \quad (47)$$

For $i = j$, using (25), all four terms collapse pairwise into $2^2 D_{ii}^{\bar{i}\bar{i}} + 2^2 D_{ii}^{\bar{i}i} = 2(-n_i/2) + 2(-n_i/2) = -2n_i$, so

$$C_{ii}^L = -2n_i. \quad (48)$$

With the $\frac{1}{2}$ prefactor and $L_{ii} = 0$ (source paper),

$$\frac{1}{2}C_{ii}^L(L_{ii} - L_{ii}^G) = -n_i(0 - L_{ii}^G) = +n_iL_{ii}^G. \quad (49)$$

This diagonal contribution, $\sum_i n_i L_{ii}^G$, is precisely the term that appears together with $2h_{ii} + J_{ii} + J_{ii}^G$ in the one-body-like sum of Eqs. (72)–(76), even though $L_{ii} = 0$ identically.

4.6 The $\Pi(K + K^G)$ block

For $i \neq j$, using (17),

$${}^2D_{ii}^{j\bar{j}} = {}^2D_{jj}^{\bar{i}i} = \frac{\Pi_{ij}}{2} \implies E_{\Pi K} = \sum_{i \neq j} \Pi_{ij}(K_{ij} + K_{ij}^G). \quad (50)$$

4.7 The $\Pi(L + L^G)$ block

For $i \neq j$, using (18),

$${}^2D_{jj}^{\bar{i}i} = {}^2D_{ii}^{j\bar{j}} = {}^2D_{jj}^{\bar{i}i} = {}^2D_{ii}^{\bar{j}j} = -\frac{\Pi_{ij}}{2} \implies -\frac{1}{2}(-4 \cdot \frac{\Pi_{ij}}{2}) = \Pi_{ij}, \quad (51)$$

so

$$E_{\Pi L} = \sum_{i \neq j} \Pi_{ij}(L_{ij} + L_{ij}^G). \quad (52)$$

Combining (50) and (52),

$$E_{\Pi} = \sum_{i \neq j} \Pi_{ij}(K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G). \quad (53)$$

5 The master formula

Collecting (37), (40), (43), (45), (47), (49), and (53), and grouping diagonal versus off-diagonal terms,

$$\begin{aligned} E = & \sum_i n_i(2h_{ii} + J_{ii} + J_{ii}^G + L_{ii}^G) \\ & + \sum_{i \neq j} (n_i n_j - \Delta_{ij}) \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right] \\ & + \sum_{i \neq j} \Pi_{ij} \left[K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G \right]. \end{aligned} \quad (54)$$

This identity holds for *any* valid choice of Δ and Π satisfying the conventions of Sec. 1; it is independent of which of the four Piris-type functionals is used.

6 Separation into intra- and inter-subspace terms

Insert the subspace ansatz (8) into (54).

Intra-subspace ($i, j \in \Omega_p$)

Here $n_i n_j - \Delta_{ij} = 0$, so the second sum in (54) drops out entirely and only Π^{intra} survives:

$$E_p = \sum_{i \in \Omega_p} n_i(2h_{ii} + J_{ii} + J_{ii}^G + L_{ii}^G) + \sum_{\substack{i, j \in \Omega_p \\ i \neq j}} \Pi_{ij}^{\text{intra}}(K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G). \quad (55)$$

This is *identical* to Eq. (73) of the source paper.

Inter-subspace ($i \in \Omega_q, j \in \Omega_p, q \neq p$)

Here $n_i n_j - \Delta_{ij} = n_i n_j$, so

$$E_{qp} = \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right] + \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} \Pi_{ij;pq}^{\text{inter}} (K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G). \quad (56)$$

This is *identical* to Eq. (75) of the source paper. Since $E = \sum_p E_p + \sum_{q \neq p} E_{qp}$, Eq. (72) is recovered exactly.

Equations (55)–(56) are still generic in $\Pi_{ij;pq}^{\text{inter}}$: they hold for *any* of the four functionals. We now specialize.

7 Recovery of each functional's final energy expression

7.1 PNOF5

Substituting $\Pi_{ij;pq}^{\text{inter}} = 0$ [Eq. (30)] into (56) kills the second sum identically:

$$E_{qp}^{\text{PNOF5}} = \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right]. \quad (57)$$

Together with the (unchanged) intra-subspace term (55) using Π^{intra} of (10), this reproduces the relativistic PNOF5 functional exactly.

7.2 PNOF7

Substituting $\Pi_{ij;pq}^{\text{inter}} = -\sqrt{n_i n_j h_i h_j}$ [Eq. (31)] into (56),

$$E_{qp}^{\text{PNOF7}} = \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right] - \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} \sqrt{n_i n_j h_i h_j} (K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G). \quad (58)$$

This is exactly the relativistic PNOF7 term of Eq. (75).

7.3 PNOF7s

Substituting $\Pi_{ij;pq}^{\text{inter}} = -4n_i n_j h_i h_j$ [Eq. (32)] into (56),

$$E_{qp}^{\text{PNOF7s}} = \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right] - 4 \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j h_i h_j (K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G). \quad (59)$$

7.4 GNOF

Substituting the piecewise $\Pi_{ij;pq}^{\text{inter, GNOF}}$ of (34) into (56) gives, with no further change to the reconstructed-2-RDM algebra,

$$E_{qp}^{\text{GNOF}} = \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} n_i n_j \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right] + \sum_{i \in \Omega_q} \sum_{j \in \Omega_p} \Pi_{ij;pq}^{\text{inter, GNOF}} (K_{ij} + K_{ij}^G + L_{ij} + L_{ij}^G), \quad (60)$$

i.e. exactly the relativistic GNOF term of Eq. (75)–(76).

In every case,

$$E^x = \sum_p E_p + \sum_{q \neq p} E_{qp}^x, \quad x \in \{\text{PNOF5, PNOF7, PNOF7s, GNOF}\}, \quad (61)$$

with E_p common [Eq. (55)] and E_{qp}^x given by (57)–(60): these are precisely Eqs. (72) and (75) of the source paper, specialized functional by functional.

8 The no-pair Dirac–Hartree–Fock limit

As a corollary, let $n_i \in \{0, 1\}$ for every i (idempotent 1-RDM). Then $h_i = 1 - n_i \in \{0, 1\}$ as well, and for *any* pair i, j (same or different subspace) either $n_i = 0$, $n_j = 0$, $h_i = 0$, or $h_j = 0$. Hence

$$\sqrt{n_i n_j} = 0 \text{ or } 1 \text{ trivially,} \quad \sqrt{n_i n_j h_i h_j} = 0, \quad n_i n_j h_i h_j = 0, \quad n_i^d n_j^d - \sqrt{n_i n_j h_i h_j} \pm \sqrt{n_i^d n_j^d} \rightarrow 0, \quad (62)$$

because in the strict single-determinant limit each subspace Ω_p contains exactly one occupied orbital ($n_i = 1$) and the rest unoccupied ($n_j = 0$); this kills Π_{ij}^{intra} as well (one of n_i, n_j vanishes for any intra-subspace pair with $i \neq j$ in this limit). Consequently every Π -dependent term in (55)–(56) vanishes for all four functionals simultaneously, leaving

$$E \rightarrow 2 \sum_i^{N_e/2} h_{ii} + \sum_{i,j}^{N_e/2} \left[2J_{ij} - (K_{ij} - K_{ij}^G) - (L_{ij} - L_{ij}^G) \right], \quad (63)$$

which is exactly the no-pair Dirac–Hartree–Fock energy, Eq. (66) of the source paper. This confirms that PNOF5, PNOF7, PNOF7s, and GNOF all collapse to the same DHF limit, as stated in Sec. 3.6 of the source paper.

9 Summary table

Functional	$\Pi_{ij;pq}^{\text{inter}}$	Explicit ${}^2D_{ii}^{jj} = {}^2D_{jj}^{ii}$
PNOF5	0	0
PNOF7	$-\sqrt{n_i n_j h_i h_j}$	$-\frac{1}{2}\sqrt{n_i n_j h_i h_j}$
PNOF7s	$-4n_i n_j h_i h_j$	$-2n_i n_j h_i h_j$
GNOF	piecewise, Eq. (34)	$\frac{1}{2}\Pi_{ij;pq}^{\text{inter, GNOF}}$

For every functional, ${}^2D_{jj}^{ii} = {}^2D_{ii}^{jj} = {}^2D_{jj}^{ii} = {}^2D_{ii}^{jj} = -{}^2D_{ii}^{jj}$, and the ordinary Coulomb/exchange/spin-flip elements (21)–(25) and (26)–(27) are common to all four.

10 Conclusion

We have written down, fully explicitly and separately for PNOF5, PNOF7, PNOF7s, and GNOF, every reconstructed 2-RDM element ${}^2D_{ij}^{kl}$ entering the relativistic no-pair natural-orbital energy expression. Substituting these explicit elements into Eq. (64) and evaluating each of its six blocks in full detail reproduces, term by term and without any additional assumption, the published energy expressions Eqs. (72)–(76) of Sec. 3.5.3 of SciPost Chem. 1, 004 (2022), and their common no-pair Dirac–Hartree–Fock limit, Eq. (66).